PRANSHU KUMAR

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EDUCATION

Veermata Jijabai Technological Institute (VJTI), Mumbai **June 2027**
Bachelor of Technology in Electronics and Telecommunications; Minor in AI & Machine Learning **9.17 CGPA**

TECHNICAL SKILLS

- **Programming:** C, C++, Python, SQL **Embedded:** nRF52840, ESP32, Zephyr RTOS, ESP-IDF, Arduino IDE, BLE, I^2C
- **ML, Vision & Data:** PyTorch, YOLOv8, OpenCV, EasyOCR, scikit-learn, NumPy, SciPy, Pandas, Excel
- **Tools & OS:** MATLAB/Simulink, ngspice, AutoCAD, Git, pytest, FastAPI, Plotly Dash; Linux, Windows, macOS

PROFESSIONAL EXPERIENCE

IIT Bombay, Mumbai, India | *IoT Research Intern – Machine Intelligence Program* **June 2026 – July 2026**
- Independently conceived, pitched and built ParkinSense, a wearable continuous-monitoring platform for Parkinson's disease, from concept to a working end-to-end prototype (firmware, BLE protocol, real-time signal processing, dashboard).
- Extended the lab's nRF52840/BLE IMU work into three parallel pipelines (motion, PPG/cardiac, activity) on one synchronised 104 Hz stream, replacing offline single-axis tremor analysis with real-time, multi-axis, motion-context-aware detection.

Scientechnic, Dubai, UAE | *Project Engineering Intern – Building Technologies Department* **June 2025 – July 2025**
- Implemented, tested and commissioned a KNX lighting control system at Amazon DXB30 Innovation Hub, Dubai Internet City: actuator configuration, addressing and line diagnostics on a central monitoring system.
- Assisted in professional AV system design (DSPs, matrix switchers, amplifiers, control processors, AV-over-IP) against venue acoustics, display sizing and lighting/BMS integration.
- Designed I/O schedules (AI, DI, AO, DO) and selected DDC controllers for BMS projects including Burj Binghatti Jacob & Co. Residences and GEMS Founders School, sized from equipment schedules, load demand and spare margins.

IIT Bombay, Mumbai, India | *IoT Research Intern – Machine Intelligence Program* **Feb 2025 – May 2025**
- Acquired high-rate IMU data on nRF52840 Sense and ESP32 and optimised BLE transfer for IoT sensor networks.
- Diagnosed EMI-driven failures on a waterproof-cable serial link for a V-Guard submerged-pump project and contributed to the move to UDP wireless transmission, improving reliability and throughput under interference; validated real-time data retrieval.

ACADEMIC PROJECTS

ParkinSense – Wearable IMU + PPG Platform for Parkinson's Tremor Monitoring nRF52840, BLE, Python, SciPy, Dash
- Built a wrist wearable (XIAO nRF52840 Sense, LSM6DS3 IMU at 104 Hz, MAX30102 PPG) streaming versioned 208-byte BLE packets of 10 samples with sequence numbers, gap detection and measured device rate; designed BLE protocol v2.
- Engineered tremor analytics: Welch PSD on 4 s windows, 3.5–7.5 Hz peak interpolation, SNR and harmonic rejection, and a 2-state HMM; raised sensitivity from 0.69 to 0.99 at 1.00 specificity with 0.024 Hz median frequency error (synthetic ground truth).
- Developed PPG heart rate and HRV (RMSSD, SDNN, pNN50) with SEARCHING/LOCKED/TRACKING/LOST lock states and motion-artefact flags; HR MAE at rest 0.37 BPM and RMSSD MAE 6.7 ms on synthetic ground truth.
- Built a motion-gated, tremor-guarded step counter that cut false steps during rest tremor from 23.3 to 0.3/min (synthetic); 69 steps against $\approx$86 expected on a real walking recording.
- Delivered a FastAPI + Dash runtime with bit-exact replay, per-recording provenance (79-column capture) and an 84-test CI suite with golden regression on real recordings.
- Firmware v4 verified on hardware; v5 (protocol v2, $\sim$100 Hz PPG) written and host-tested.

CircuitVision – Hand-Drawn Circuit to SPICE Netlist (team of 2, AI/ML minor) YOLOv8, OpenCV, EasyOCR, ngspice
- Trained YOLOv8s on the CGHD handwritten-circuit dataset (2,981 photos, 9 classes), reaching 0.904 mAP50 and 0.611 mAP50-95 on an unseen drafter.
- Built paper-relative binarisation, HOG + SVM orientation and polarity classification, and wire tracing with symbol masking and

crossovers (0.994 exact topology across 11 synthetic photo conditions).

- Generated SPICE netlists from EasyOCR-read values, validated by a CFG/Earley parser and simulated in ngspice; added an HMM (Viterbi) text-to-netlist path scoring 0.90 exact on held-out sentences; 136 automated tests.
- Measured honestly on real photos: correct circuit on 37.5% of 48 analog photos with ground-truth boxes and 10.4% with detector boxes; OCR on real handwriting remains the main bottleneck.

TremoSense – Tremor-Suppressing Spoon on Zephyr RTOS (team of 3) Zephyr, C, EKF, LQR, Simulink

- Built a two-axis servo gimbal (MG995 roll, FS90MG pitch) on the XIAO nRF52840 Sense: 208 Hz IMU, 2-state EKF roll/pitch estimation and a 2×4 LQR gain, with servo commands every 10 ms through Zephyr threads, a message queue and a semaphore.
- Validated the EKF–LQR design in MATLAB/Simulink at 1000 Hz: over 90% roll and 80% pitch suppression across most of the 3–7 Hz tremor band; residual RMS 0.346°/0.573° under 8°, 8 Hz tremor.

NaviSight – Assistive Navigation for the Visually Impaired Arduino, ESP32-CAM, TensorFlow Lite

- Built an ultrasonic smart cane with haptic and buzzer alerts and GPS (L80-M39) + HC-05 location sharing by SMS, and deployed offline TensorFlow Lite object recognition on an ESP32-CAM.

Wall-E – Self-Balancing, Line-Following Robot (SRA team) ESP32, ESP-IDF, PID

- Implemented PID line following from a Light Sensor Array and PID self-balancing on an ESP32 with the in-house SRA Board; tuned gains through iterative testing.

ACHIEVEMENTS & EXTRA-CURRICULAR ACTIVITIES

- Academic Branch Topper (First Year) · Organising team, Technovanza 2024 (4,000–5,000 participants) · Member, Society of Robotics and Automation, VJTI · Guitarist at college cultural programmes